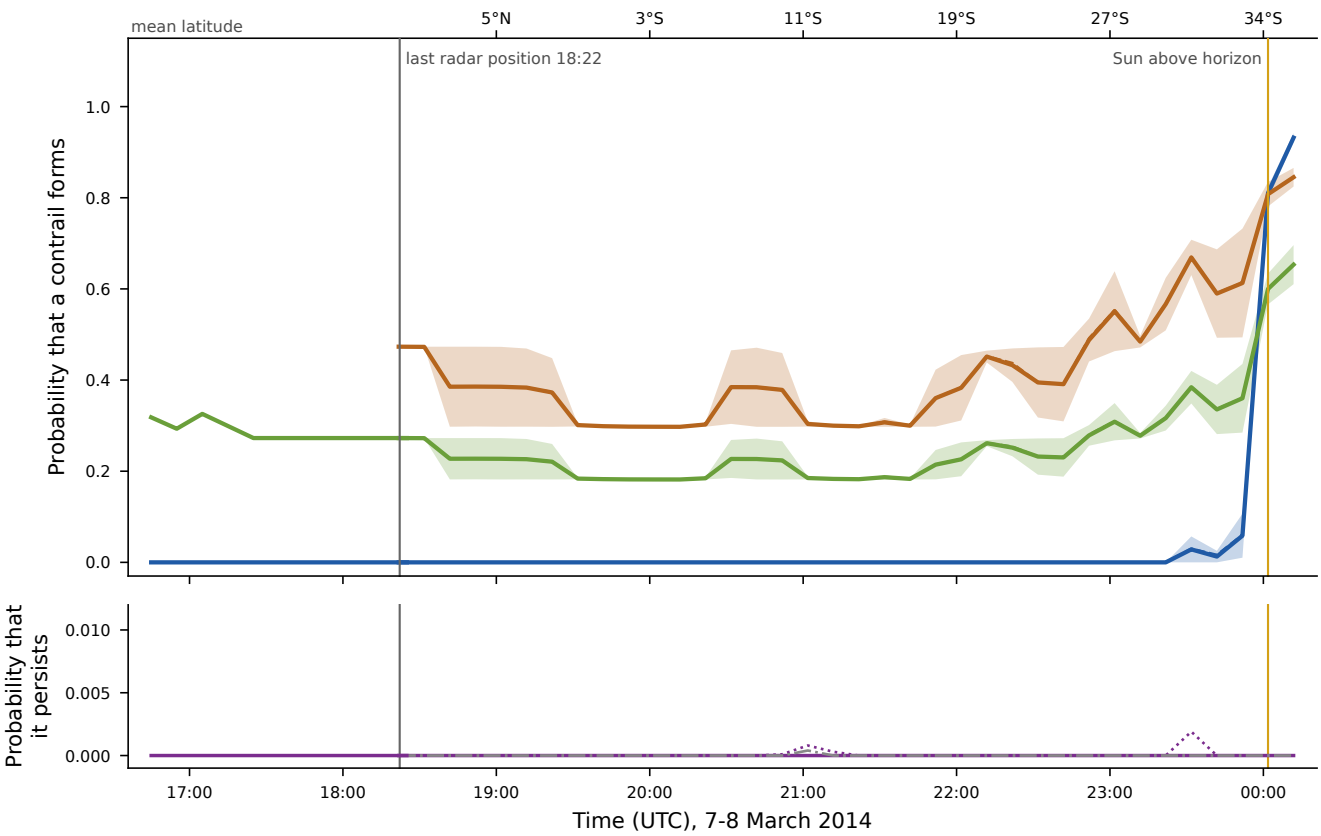
a

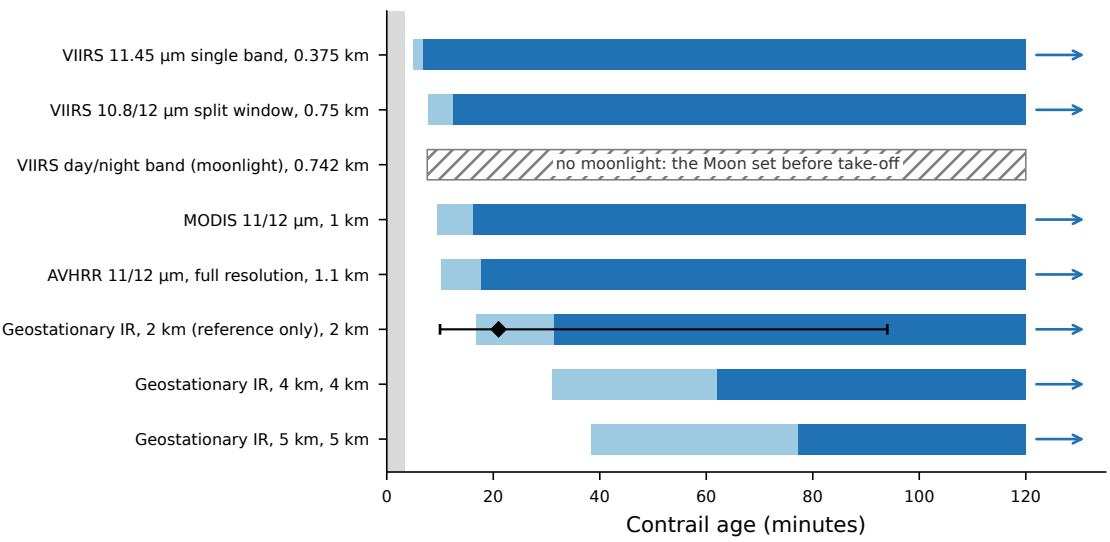
A contrail could form at the highest flight levels, but it could not persist



- Formation at FL350, humidity corrected
- Formation at FL350, raw ERA5 humidity
- Formation, each level FL300-FL400 equally likely, humidity corrected
- Formation, each level FL300-FL400 equally likely, raw ERA5 humidity
- Formation, posterior altitude distribution (from 18:22), humidity corrected
- Formation, posterior altitude distribution (from 18:22), raw ERA5 humidity
- Persistence, all altitude cases, raw and humidity corrected
- Persistence, stress test, humidity divided by 0.9 (levels equally likely)
- Persistence, stress test, humidity divided by 0.8 (levels equally likely)
- Persistence, stress test, temperature 1 K lower (levels equally likely)

b

If the contrail had persisted, polar-orbiter images could show it after 5-18 minutes, and geostationary images after 31-77 minutes



- First 200 s: contrail 0.08-0.15 km wide, narrower than every pixel
- Onset depends on the spreading rate (2.3 m/s observed mean, down to the ERA5 wind-shear rate)
- Detectable in ice-supersaturated air; typical end about 2 h (arrow: can last several hours)
- Possible in principle, but no illumination on this night
- Median and quartiles of first detection, simulated 2 km imager (Driver et al. 2025)

Plain-language note: This chart shows the chance that the aircraft made a contrail on the possible routes. At FL300 to FL360 the air was too warm, except near the end of the flight. Higher up, a contrail could form, but the air was too dry for it to stay. A contrail that does not stay goes away in a few minutes, before it is wide enough for a satellite to see. Panel b shows when satellites could see a contrail if it had stayed. On this night a contrail would not have stayed long enough for these satellites to see it.

Technical note: ERA5 hourly 0.25° pressure-level temperature and specific humidity (ARCO-ERA5 public mirror), interpolated linearly in time, bilinearly in space and linearly in ln p to ISA flight-level pressures (FL300 300.9 hPa, FL350 238.4 hPa, FL400 187.5 hPa). Routes: run J posterior, 148,000 routes (74 seeds x 2,000), family weights waypoints 0.9982, free 0.0013, radar reproduction 0.0003, descent-climb 0.0002; 10-min vertices 18:21:49-00:11:49 UTC. Before 18:22: Geoscience Australia digital radar track (altitude known to 17:20, FL350). Schmidt-Appleman criterion (Schumann 1996): water emission index 1.25 kg/kg, fuel heat 43.2 MJ/kg, overall propulsion efficiency 0.30-0.35 (shaded band; line = mean), Sonntag (1994) saturation pressures. Humidity corrected = Teoh et al. (2024) latitude-dependent ERA5 correction (pycontrails coefficients); Teoh et al. (2022) gives the same persistence. Persistence = criterion met and RHi >= 100 %; posterior-weighted maximum < 0.0001 at every level, time and correction. Posterior altitude = core trace altitudes at 18:42:40 (5/50/95 % 29,000/37,000/40,000 ft), held constant and assumed independent of route. Panel b: contrail width 0.15 km at 200 s, then linear growth at 2.3 m/s (Freudenthaler et al. 1995) or ERA5 median FL350 shear 0.0044 1/s x 250 m depth (1.09 m/s); first detection when width >= nominal nadir pixel and optical depth >= 0.05 (Driver et al. 2025); typical end of detection about 2 h (Vazquez-Navarro et al. 2015; Driver et al. 2025). Off-nadir pixels are about 2x larger at scan edge, which delays onset. Moon 43 % illuminated and below the horizon at every route point from take-off (JPL DE421).